SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 3 **Duration : 45 Minutes**
Total Marks:30

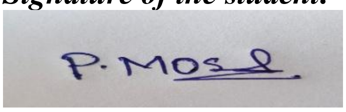
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I P.MOSHIYA(192572142).(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift
iv. D3 cannot work Morning
v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Problem Understanding	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Constraint Analysis	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Solution Development	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Application of Concepts	15% (1.5)	/1.5	/1.5	/1.5	/4.5
Justification	10% (1)	/1	/1	/1	/3
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/30**

100

1. Hospital Doctor Shift Assignment Using Backtracking Search

Problem Definition
Doctors: D1, D2, D3
Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night
   → D1 ≠ N
2. D2 must work before D3
   → Position(D2) < Position(D3)
3. Only one doctor per shift
4. D3 cannot work Morning
   → D3 ≠ M
5. Every doctor gets exactly one shift

Backtracking Search Process
Backtracking assigns values one by one and checks constraints after every assignment.
Step 1: Assign D1
Possible shifts for D1:

- Morning ✓
- Afternoon ✓
- Night ✗ (constraint violation)

Choose:
D1 = Morning
Current assignment:

Doctor	Shift
D1	Morning

Constraint Check:

- D1 not Night → ✓ Valid

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Try:
D2 = Afternoon
Current assignment:

Doctor	Shift
D1	Morning
D2	Afternoon

Constraint Check:
D2 must work before D3.
D3 must come after Afternoon → D3 must be Night.
Possible? Yes.
Continue.

Step 3: Assign D3

Remaining shift:
Night
Assignment:

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Checks:

Constraint 1:
D1 cannot work Night
D1 = Morning ✓

Constraint 2:
D2 before D3
Afternoon < Night ✓

Constraint 3:
One doctor per shift ✓

Constraint 4:
D3 cannot work Morning
D3 = Night ✓

Constraint 5:
Every doctor assigned exactly one shift ✓

Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2

Shift	Doctor
Night	D3

Final Solution:
D1 → Morning
D2 → Afternoon
D3 → Night

2. BFS Search Algorithm for Robot Grid Problem

Since the grid is not provided, assume the following 5×5 environment:
Legend:

- S = Start
- G = Goal
- X = Obstacle
- . = Empty cell

```
    1  2  3  4  5

1   S  .  X  .  .

2   .  .  X  .  .

3   .  .  .  .  .

4   X  .  X  .  .

5   .  .  .  .  G
```

Start position:
S = (1,1)
Goal:
G = (5,5)
Movement:

- Up
- Down
- Left
- Right

Cost per move = 1

BFS Algorithm

BFS uses a queue and expands nodes level by level.
Initial Queue:
Queue = [(1,1)]

Step 1: Expand Start Node

Current node:
(1,1)
Possible moves:

Move	Result
Right	(1,2) ✓
Down	(2,1) ✓
Left	Outside

Move	Result
Up	Outside

Queue update:
[(1,2),(2,1)]

Step 2: Expand (1,2)
Possible moves:

Node	Status
(1,3)	Obstacle ✗
(2,2)	Valid ✓

Queue:
[(2,1),(2,2)]

Step 3: Expand (2,1)
Possible moves:

Node	Status
(3,1)	Valid
(2,2)	Already visited

Queue:
[(2,2),(3,1)]

Step 4: Expand (2,2)
Possible moves:

Node	Status
(3,2)	Valid
(2,3)	Obstacle
(1,2)	Visited

Queue:
[(3,1),(3,2)]

Step 5: Expand (3,1)
Possible move:
(3,2) already discovered

Queue:
[(3,2)]

Step 6: Expand (3,2)
Possible moves:

Node	Status
(3,3)	Valid
(4,2)	Valid

Queue:
[(3,3),(4,2)]

Step 7: Expand (3,3)

Possible moves:

Node	Status
(3,4)	Valid
(4,3)	Obstacle
(2,3)	Obstacle

Queue:
[(4,2),(3,4)]

Step 8: Expand (4,2)

Possible moves:
(5,2)

Queue:
[(3,4),(5,2)]

Step 9: Expand (3,4)

Possible moves:
(4,4), (2,4)

Queue:
[(5,2),(4,4),(2,4)]

Step 10: Continue BFS

Path reaches:
(5,5)
Goal found.

Optimal Path

The shortest path found by BFS:
S
↓
(2,1)
↓
(3,1)
↓
(3,2)
↓
(3,3)
↓
(3,4)
↓
(4,4)
↓
(5,4)
↓
G

Path Cost

Number of movements:
= 8 moves
Therefore:

Optimal Cost = 8
BFS guarantees the shortest path because every move has equal cost.

3.Autonomous Rescue Robot AI Problem

(a) Constraint Analysis and Effect on Decision Making

The rescue robot operates in a partially observable environment.

1. Movement Constraints

Allowed movements:

- Up
- Down
- Left
- Right

Effect:

- Robot must select only valid neighboring cells.
- Diagonal movement is not allowed.

2. Obstacles

Blocked areas cannot be entered.
Effect:

- Robot must detect blocked paths.
- It must search alternative routes.

3. Limited Sensing Capability

Robot can only observe adjacent cells.
Effect:

- The robot does not know the complete map.
- It must gather information while moving.
- Decisions are made based on current knowledge.

4. Movement Cost

Normal movement:
Cost = 1
Risk zone:
Additional cost = 2
Therefore:
Risk zone movement cost:
1 + 2 = 3
Effect:

- Robot avoids dangerous areas if possible.
- The cheapest route is preferred over the shortest route.

5. Dynamic Knowledge Update

The robot learns new information during exploration.
Effect:

- The internal map changes continuously.
- Previously planned routes may need modification.

(b) Solution Approach Using Uniform Cost Search / Online Search Agent

Selected Method:
Online Uniform Cost Search (UCS)

Reason:

- Movement costs are different.
- Risk areas have higher cost.
- UCS finds the minimum-cost path.

Steps of Algorithm

Step 1: Initial State

Robot starts at S.

Knowledge:

Known location = S
Cost = 0

Step 2: Sense Environment

Robot observes nearby cells:

- Open paths
- Obstacles
- Risk zones

Step 3: Generate Possible Actions

Available actions:

- Move Up
- Move Down
- Move Left
- Move Right

Step 4: Calculate Path Cost

For every possible movement:
Normal cell:
Total cost = Current cost + 1
Risk cell:
Total cost = Current cost + 3

Step 5: Select Lowest Cost Node

UCS selects the node with minimum accumulated cost.
Priority Queue example:

Node	Cost
A	2
B	3
C	5

Robot chooses A.

Step 6: Update Knowledge

After moving:

- Update map
- Remove impossible paths
- Add newly discovered paths

Step 7: Continue Until Goal

Repeat until:

Robot reaches survivor location G.

(c) AI Concepts Used

1. Intelligent Agent

The rescue robot is an intelligent agent because it:

- Observes the environment using sensors.
- Makes decisions.
- Performs actions.
- Learns from new information.

Agent cycle:

Perceive → Think → Act → Learn

2. Rationality

A rational robot chooses actions that maximize success.

The robot:

- Avoids obstacles.
- Avoids dangerous zones.
- Minimizes travel cost.
- Reaches survivors safely.

The rational decision is not always the shortest path, but the safest and lowest-cost path.

3. Environment Type

The rescue environment is:

Property	Description
Partially Observable	Robot cannot see the entire building
Dynamic	Conditions may change
Sequential	Each action affects future decisions
Stochastic	Unexpected events may occur
Unknown	Complete map is unavailable
Single Agent	Only robot makes decisions

(d) Justification of Chosen Approach

Why Online Uniform Cost Search is Suitable?

1. Handles Different Costs

Normal paths cost 1, risky paths cost 3.

UCS considers these differences.

2. Works in Unknown Environments

The robot does not need a complete map initially.

It discovers information while moving.

3. Finds Minimum-Cost Solution

UCS expands the lowest-cost path first.

Therefore, it avoids unnecessary danger and expense.

4. Supports Dynamic Updates

When new obstacles appear:

- Robot updates knowledge.
- Recalculates the route.